

FREDERIKSHAVN, Denmark

WMO: 060430

Lat: 57.4361N

Lon: 10.5489E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 91-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.8	22.5	10.3	9.3	25.9	13.1	10.7	27.3	36.7	39.5	33.4	40.1	13.4	40	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.9	72.3	64.1	70.3	62.9	68.3	61.7	66.5	70.0	64.8	68.5	62.9	67.0	8.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	92.7	68.7	63.1	86.6	67.2	61.1	80.4	65.5	31.1	70.2	29.8	68.6	28.4	67.2	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.4	27.9	25.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			15.7	67.9	4.5	2.6	12.5	69.8	9.9	71.4	7.4	72.9	4.1	74.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	34.0	32.8	36.2	42.6	50.7	55.4	59.8	61.0	55.0	46.9	39.4	35.1
	DBStd	11.20	5.68	6.11	4.47	5.21	5.30	4.44	4.58	4.81	4.02	5.06	4.95	5.90
	HDD50	2604	497	482	427	231	56	4	0	0	6	121	319	461
	HDD65	7033	962	902	892	671	444	290	171	145	301	560	769	926
	CDD50	1078	0	0	0	10	77	165	305	341	155	25	0	0
	CDD65	34	0	0	0	0	0	2	11	21	0	0	0	0
	CDH74	21	0	0	0	0	0	2	5	14	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.0	15.0	14.5	13.8	11.3	11.6	12.9	11.3	11.3	12.3	13.7	14.2	13.8
Precipitation	PrecAvg	28.3	2.4	1.7	1.5	1.5	1.9	2.8	2.5	3.0	2.9	3.1	2.4	2.4
	PrecMax	38.4	4.4	3.2	3.8	3.7	3.8	5.9	6.0	5.4	10.5	5.7	4.4	4.7
	PrecMin	19.2	0.4	0.4	0.2	0.4	0.2	0.0	0.3	0.2	1.4	0.4	0.7	0.6
	PrecStd	4.6	1.1	0.9	0.9	0.9	0.9	1.4	1.5	1.3	1.8	1.4	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	48.0	50.0	62.3	67.2	70.9	74.6	75.9	68.8	59.0	52.4	48.6
		MCWB	44.9	44.2	42.1	53.0	56.2	61.5	65.2	63.5	61.5	55.0	49.8	46.7
	2%	DB	44.7	44.9	46.3	56.6	63.9	68.1	72.1	72.9	65.6	56.4	50.0	46.6
		MCWB	42.9	42.7	42.1	49.7	54.4	59.2	64.1	64.8	59.3	53.5	47.6	44.8
	5%	DB	43.1	42.9	44.2	53.0	61.6	65.5	70.0	70.9	62.9	55.1	48.3	45.3
		MCWB	41.4	40.8	40.9	47.2	54.0	57.8	63.3	63.5	57.8	52.7	46.0	43.8
	10%	DB	41.6	41.3	42.2	49.8	59.0	62.9	67.9	69.0	60.7	53.7	46.5	43.5
		MCWB	39.8	39.2	39.6	45.4	52.8	56.2	61.6	62.6	56.6	51.4	44.4	42.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.0	45.8	45.2	54.3	59.0	64.1	69.3	68.4	63.3	56.4	50.9	47.3
		MCDB	47.1	47.3	46.7	60.7	63.5	68.8	72.5	72.1	66.8	57.9	52.1	48.1
	2%	WB	43.4	43.2	43.3	50.8	57.3	61.2	67.0	66.6	61.0	54.5	48.2	45.1
		MCDB	44.3	44.3	45.1	54.7	61.3	65.9	70.1	70.2	64.5	55.8	49.5	46.1
	5%	WB	41.8	41.3	41.8	48.2	55.5	59.2	64.7	65.3	58.9	53.4	46.5	43.9
		MCDB	42.9	42.5	43.5	51.6	60.4	63.5	68.4	68.9	61.5	54.8	47.9	45.1
	10%	WB	40.1	39.5	40.3	46.2	53.5	57.5	62.5	63.8	57.4	52.0	44.8	42.3
		MCDB	41.4	41.0	42.1	49.1	57.7	61.7	66.2	67.8	59.7	53.5	46.3	43.6
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.1	6.9	8.2	9.7	10.6	10.9	9.9	7.3	6.4	5.3	5.5
		MCDBR	7.0	7.8	10.4	11.9	12.8	13.1	12.7	10.9	10.5	7.3	7.4	6.9
		MCWBR	7.0	6.5	7.4	6.2	7.7	7.5	7.1	6.9	6.3	5.7	7.1	6.7
	5% WB	MCDBR	6.7	7.4	9.0	11.0	11.2	11.5	10.3	9.7	8.6	6.8	6.8	7.0
		MCWBR	7.0	7.2	7.4	6.4	7.7	8.5	7.5	7.4	6.0	5.9	6.9	6.8
Clear-Sky Solar Irradiance	taub		0.306	0.332	0.349	0.387	0.379	0.369	0.390	0.391	0.362	0.347	0.323	0.291
	taud		2.245	2.305	2.344	2.294	2.386	2.440	2.401	2.401	2.467	2.418	2.298	2.205
	Ebn at Noon		174	224	254	261	272	276	267	258	249	217	166	143
	Edh at Noon		16	23	29	36	35	33	34	32	26	21	15	12
All-Sky Solar Radiation	RadAvg		116	294	757	1288	1719	1905	1749	1364	925	428	156	81
	RadStd		24	53	71	147	137	137	161	89	89	56	26	9

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (52 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	-216	-297	N/A	N/A

Nomenclature: See separate page